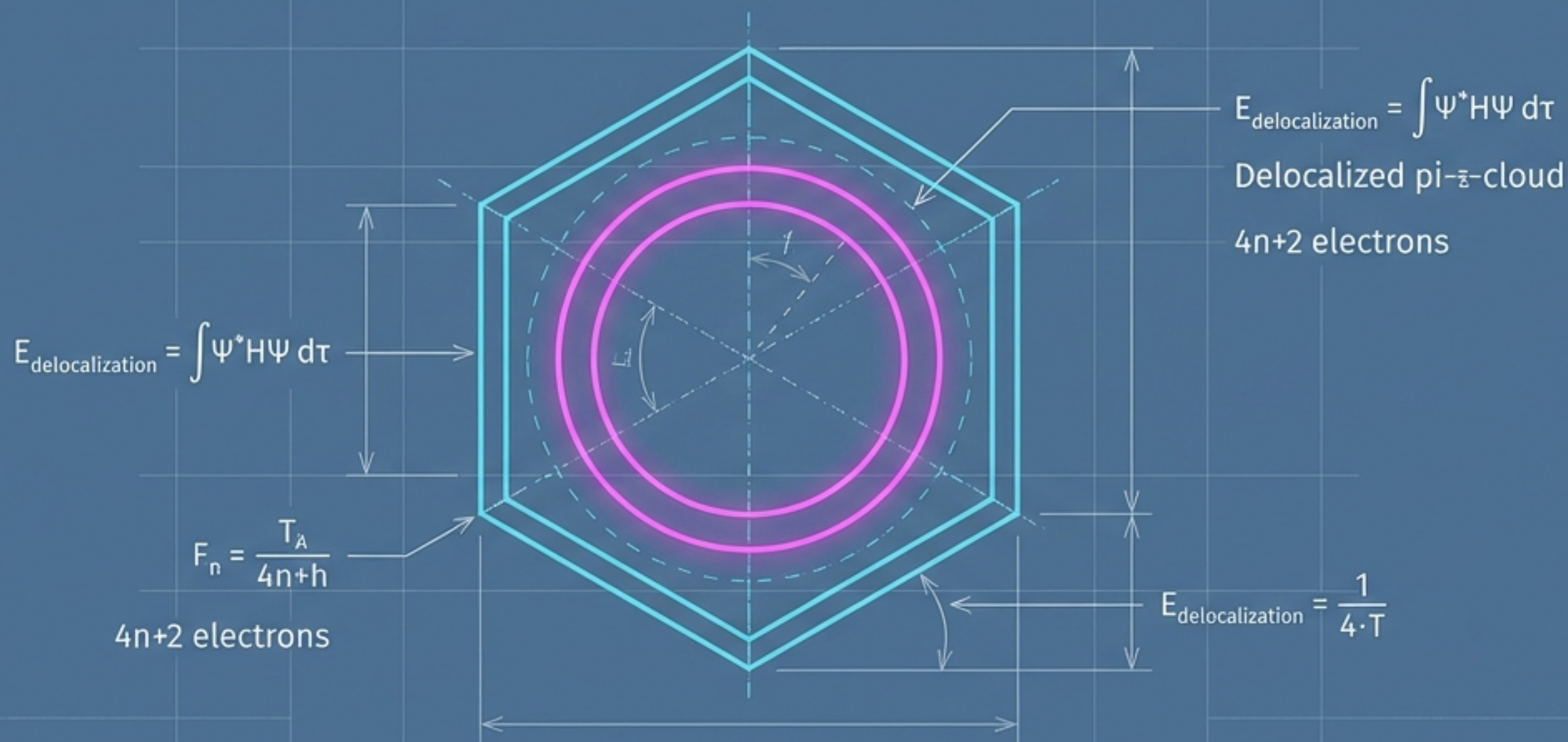
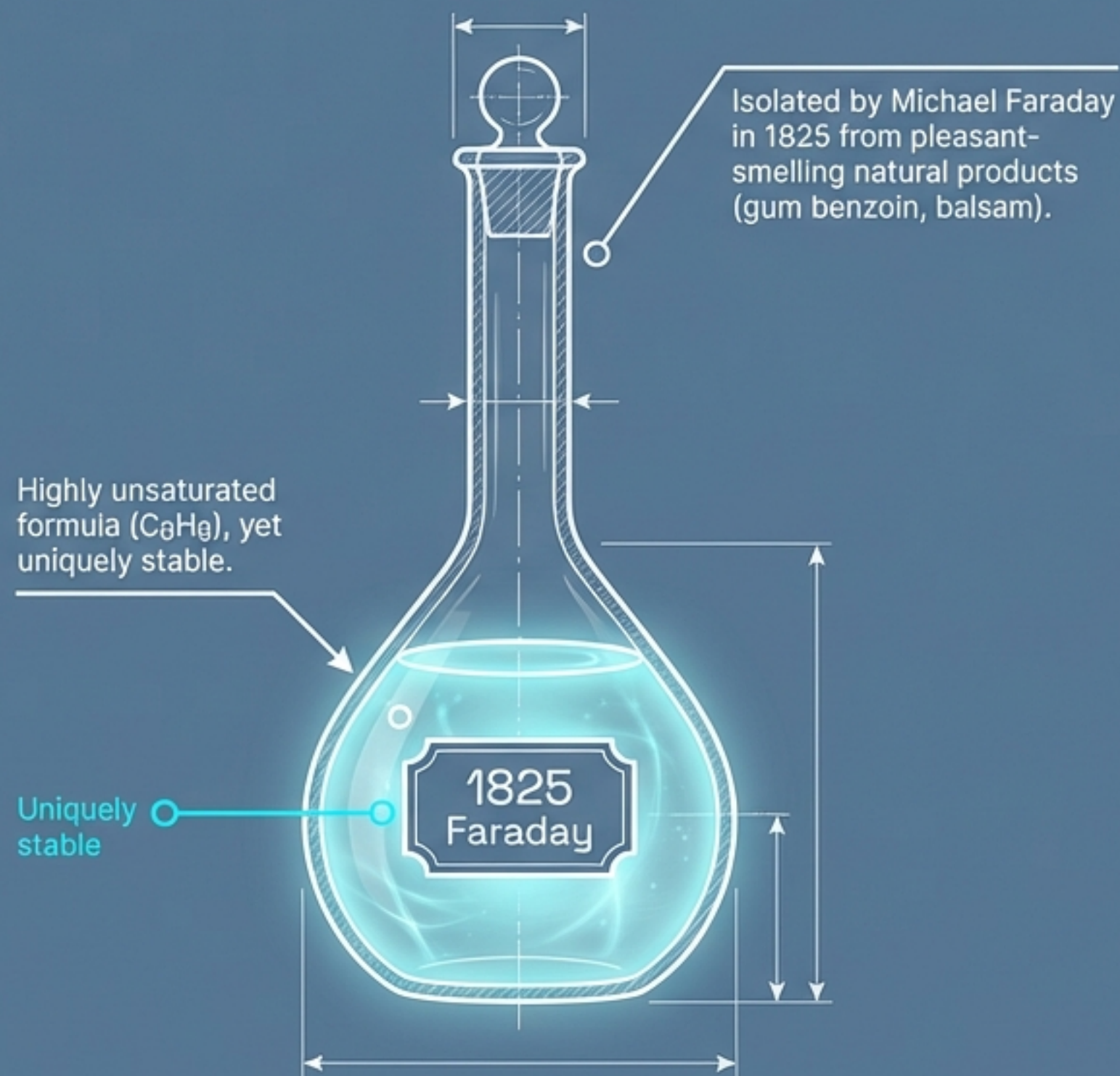


The Architecture of Aromaticity

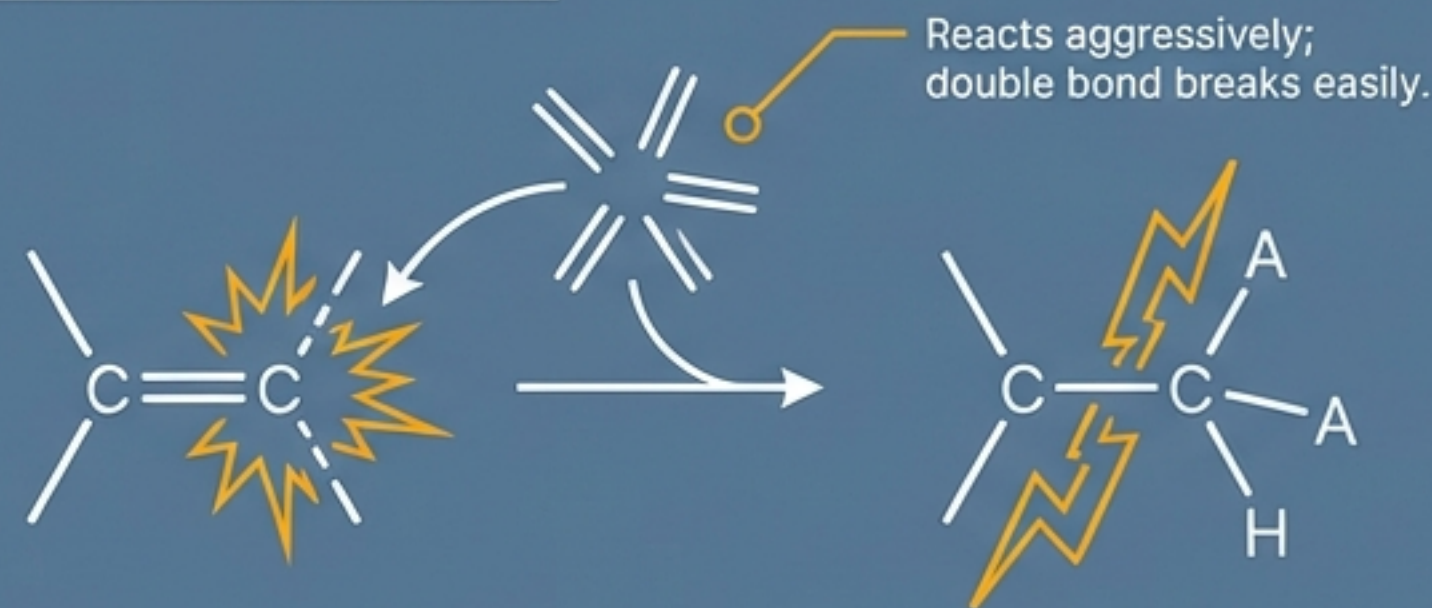
From Historical Anomalies to Quantum Blueprints



The C₆H₆ Anomaly



TYPICAL ALKENE ADDITION

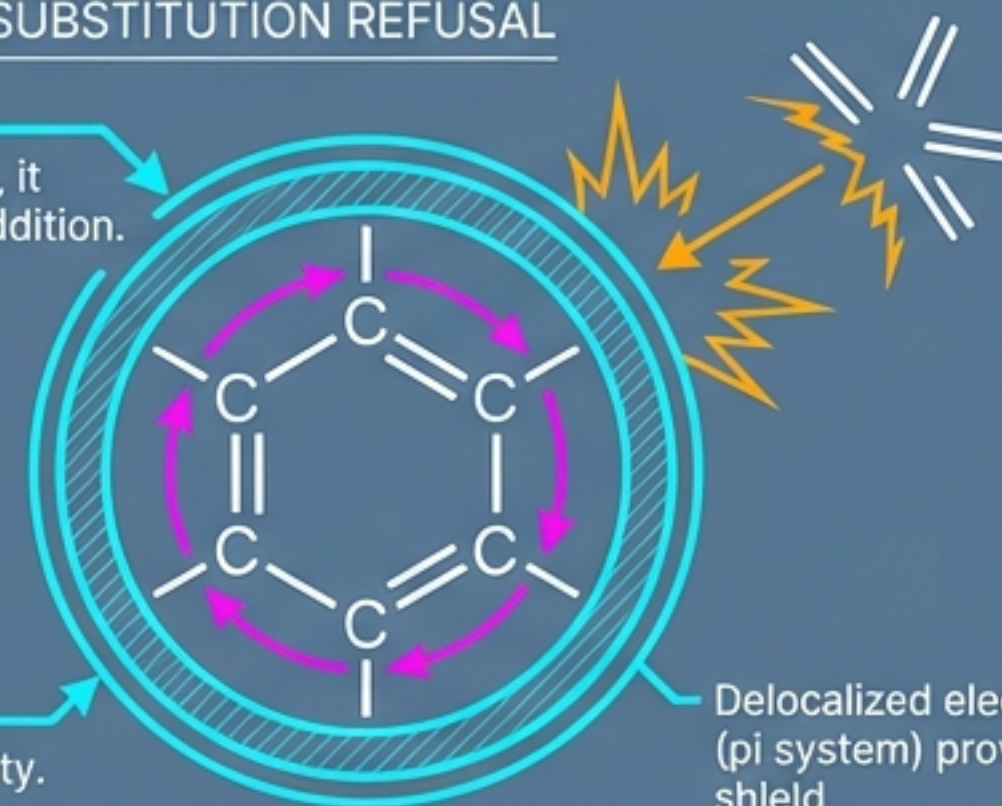


BENZENE (C₆H₆) SUBSTITUTION REFUSAL

Unlike typical alkenes, it refuses to react via addition. It protects its core.

Aromatic stability (C₆H₆ ring) > Reactivity.

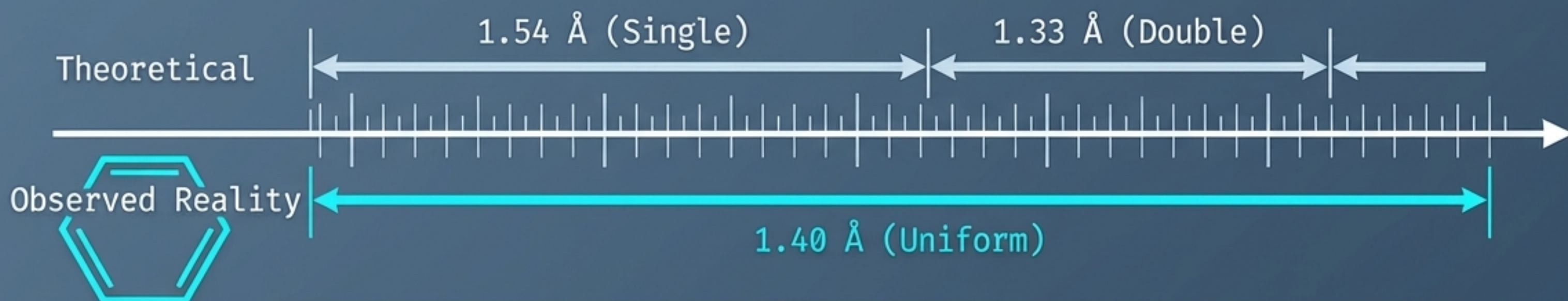
Delocalized electrons (pi system) provide shield.



The Failed Hypothesis of Oscillating Bonds

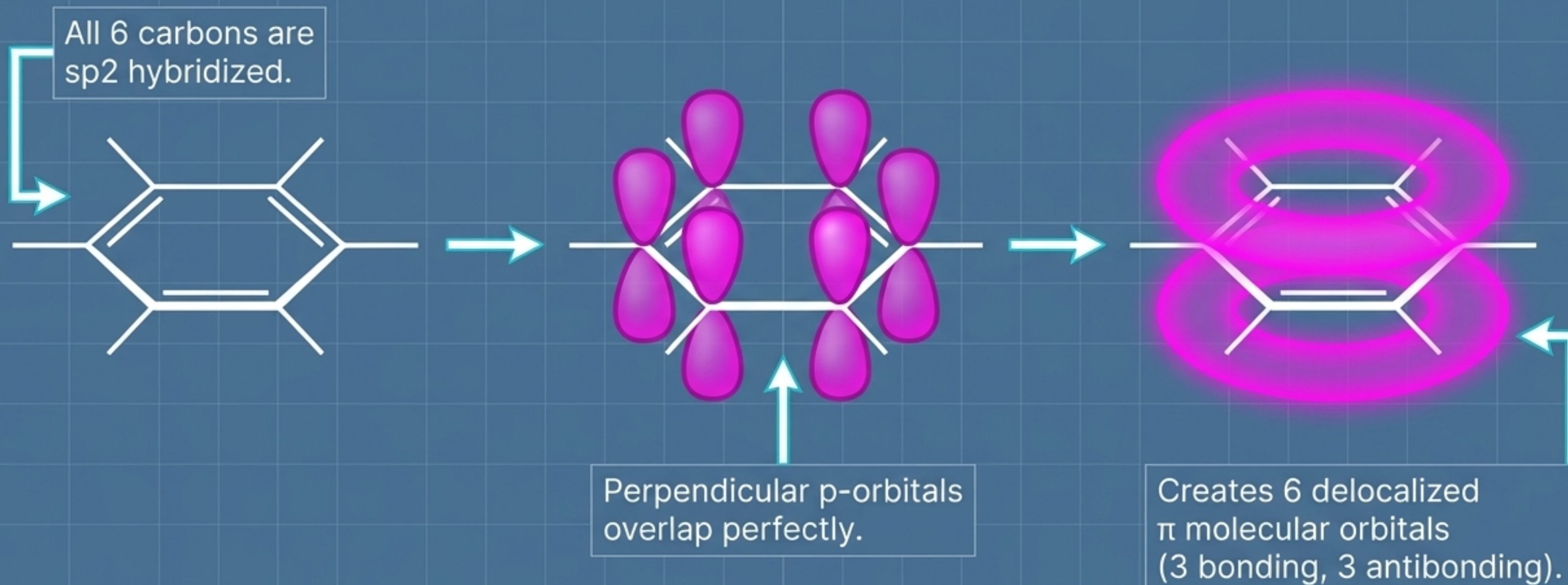


1865: Kekule proposes rapid oscillation between two structural forms in equilibrium.

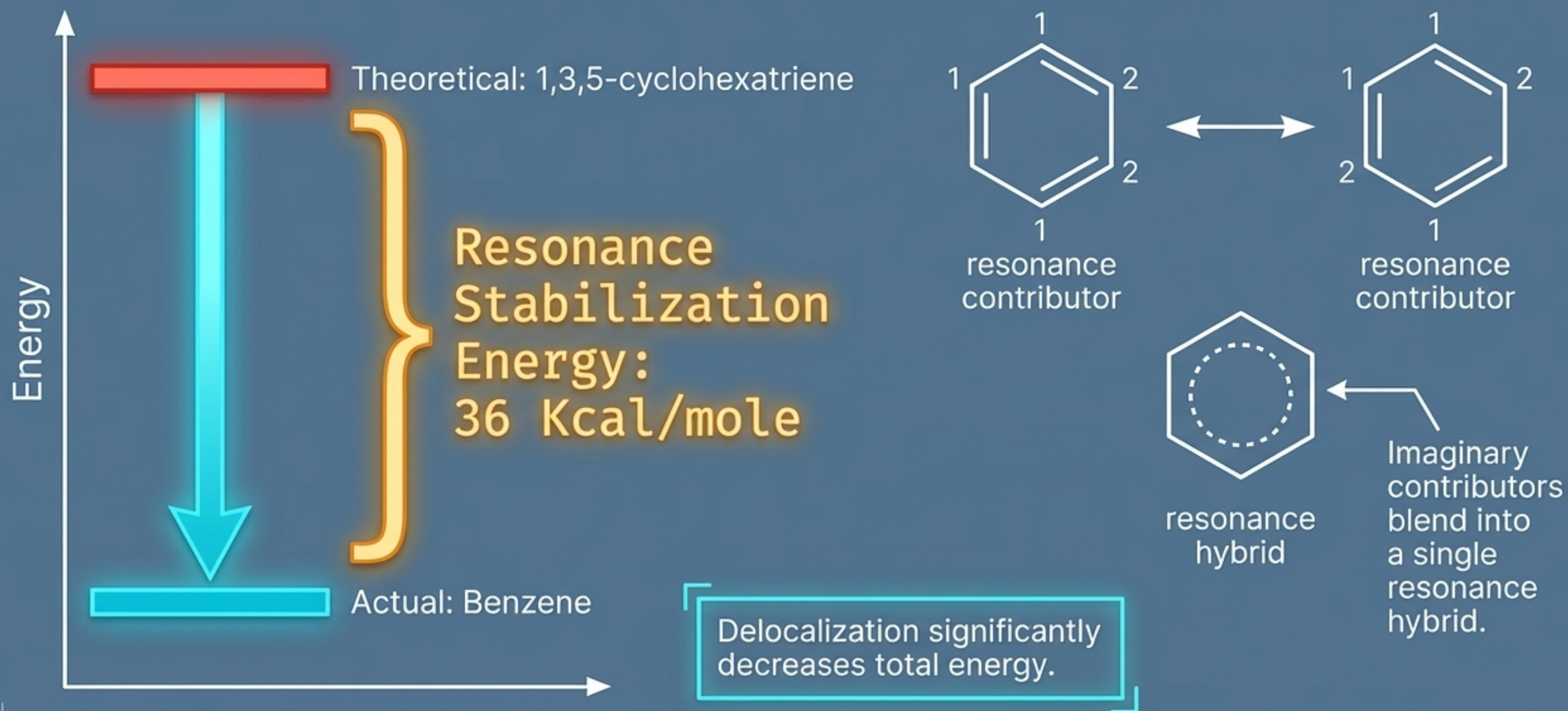


The Flaw: Fails to explain uniform 1.4 Å bond lengths.
The Flaw: Fails to explain the extreme thermodynamic stability.

The True Architecture: Molecular Orbital Theory



The 36 Kcal/mole Resonance Advantage



The Four Laws of the Ring



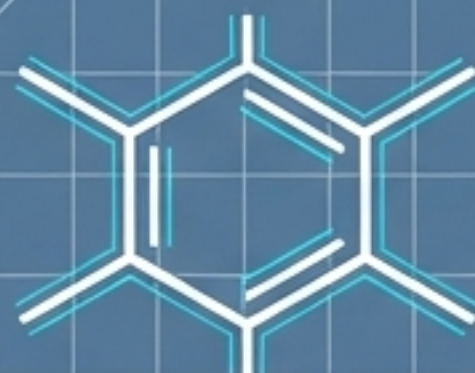
1. Cyclic

Must form a closed ring.



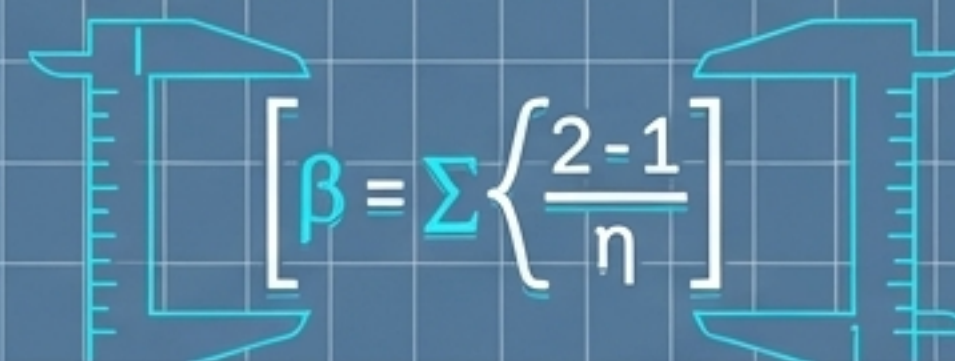
2. Planar

Must lie flat to allow orbital overlap.



3. Fully Conjugated

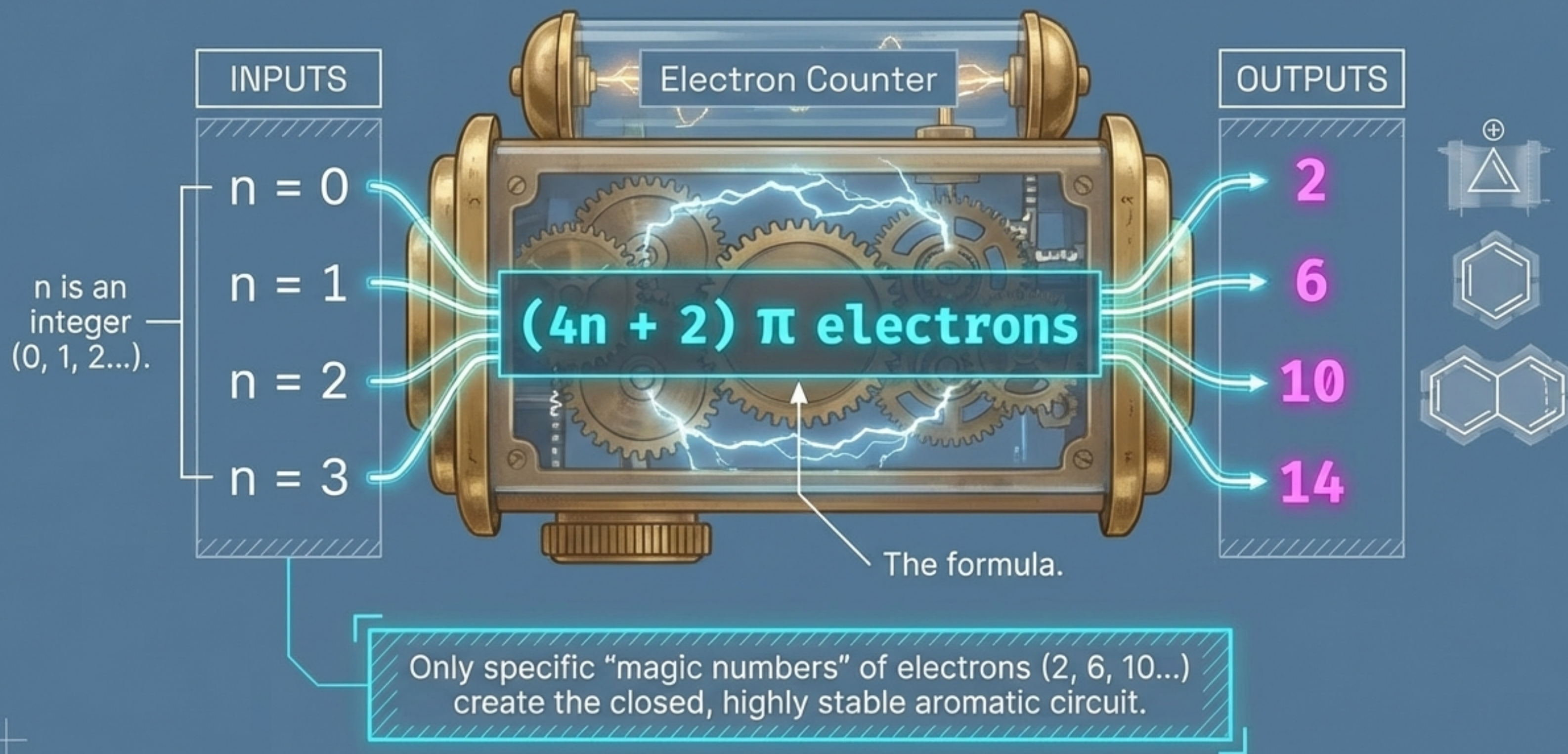
Every ring atom must be sp^2 hybridized (no sp^3 roadblocks).



4. Huckel's Rule

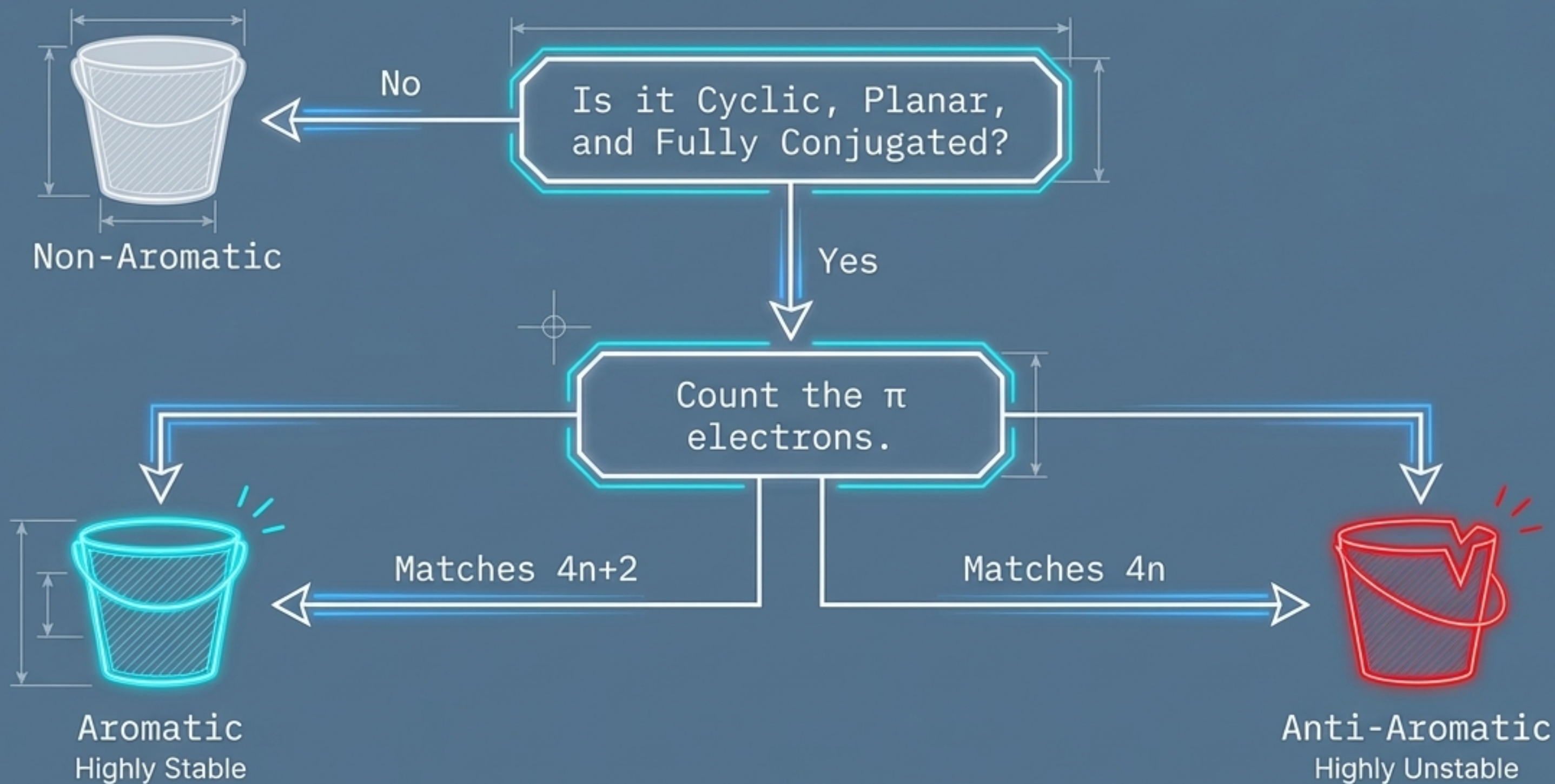
The pi-electron cloud must contain an exact mathematical quantity.

The Strict Math of Huckel's Rule

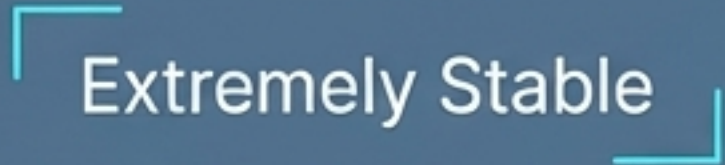
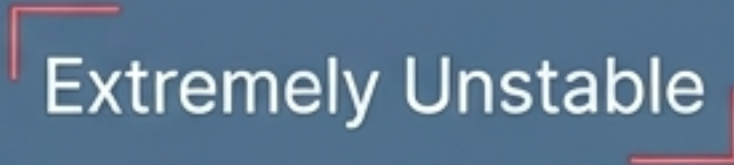


The Classification Decision Tree

Follow the structural logic to determine a molecule's chemical destiny.



The Aromaticity Spectrum

	Aromatic	Anti-Aromatic	Non-Aromatic
Criteria Met (1-3)			 (Fails rule 1, 2, or 3)
Electron Count	$4n + 2$	$4n$	N/A
Energy Level	 Extremely Stable	 Extremely Unstable	Normal / Neutral
Example			

Anti-aromatic compounds meet rules 1-3, but possess a destructive $4n$ electron count, rendering them highly reactive.

Case Study: The Cyclopentadiene Family

Neutral



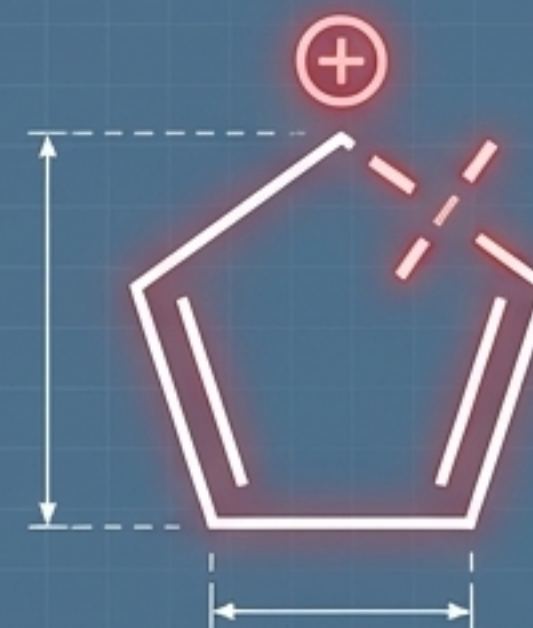
Contains an sp^3 carbon.
Fails conjugation.
Result: **Non-aromatic**

Anion



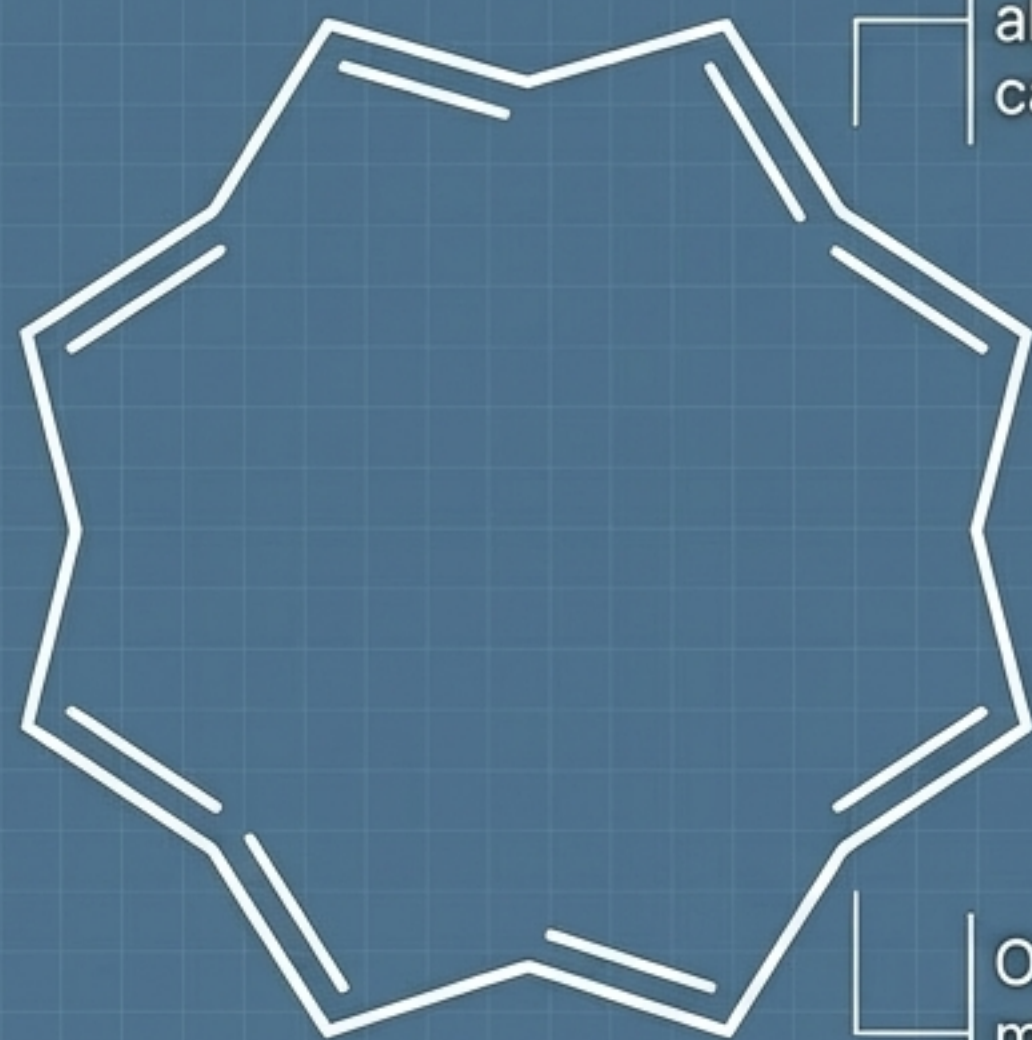
Gains a lone pair.
 6π electrons ($n=1$).
Result: **Highly stable Aromatic**
(Explains why neutral cyclopentadiene is unusually acidic—it wants to lose H^+ to become aromatic).

Cation



4π electrons ($n=1/2$).
Math fails.
Result: **Anti-aromatic**

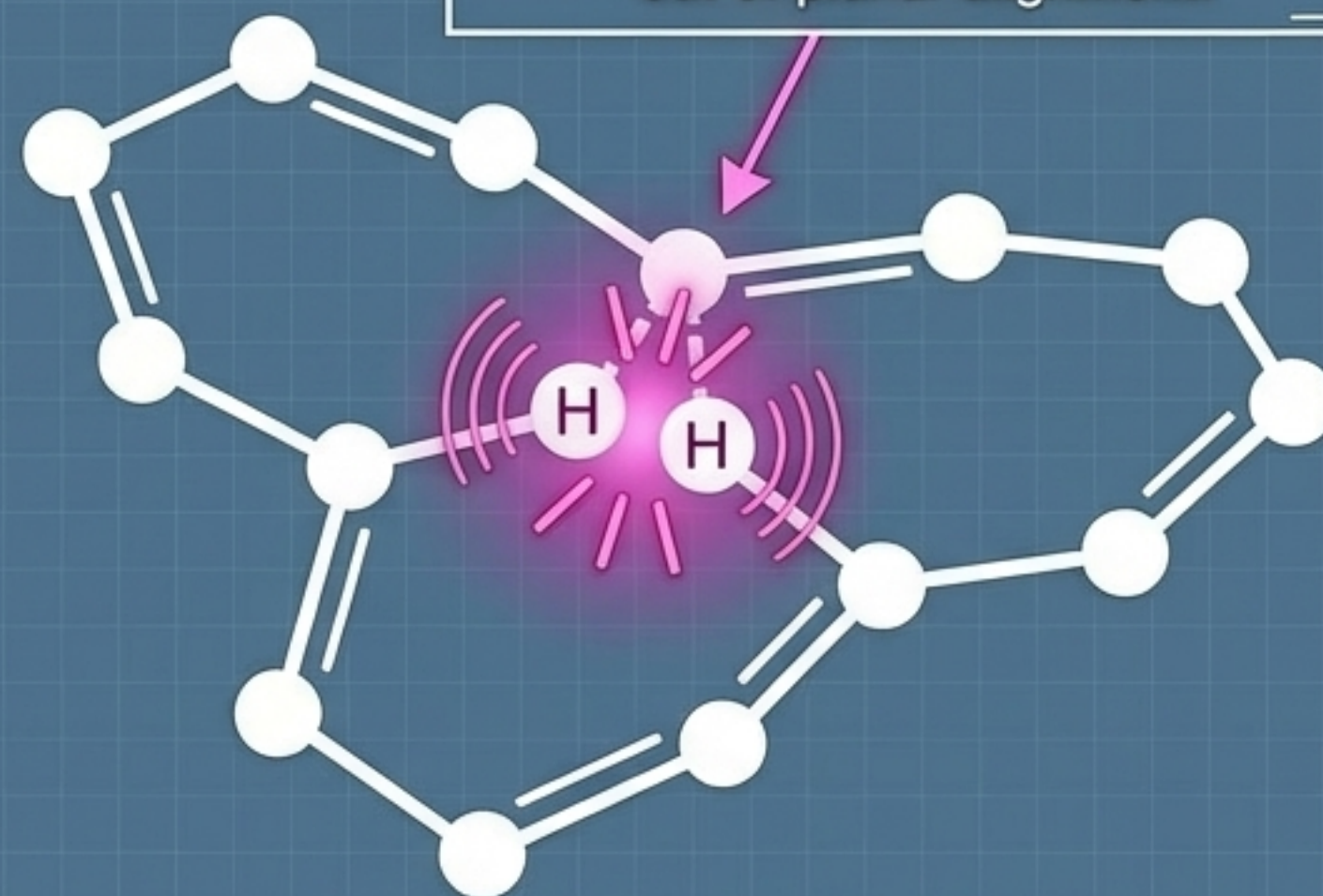
Edge Cases: Annulenes and the Planarity Trap



[10]Annulene

Annulenes: Cyclic alkenes named by carbon count.

Obeys Huckel's math exactly: 10π electrons ($n=2$).



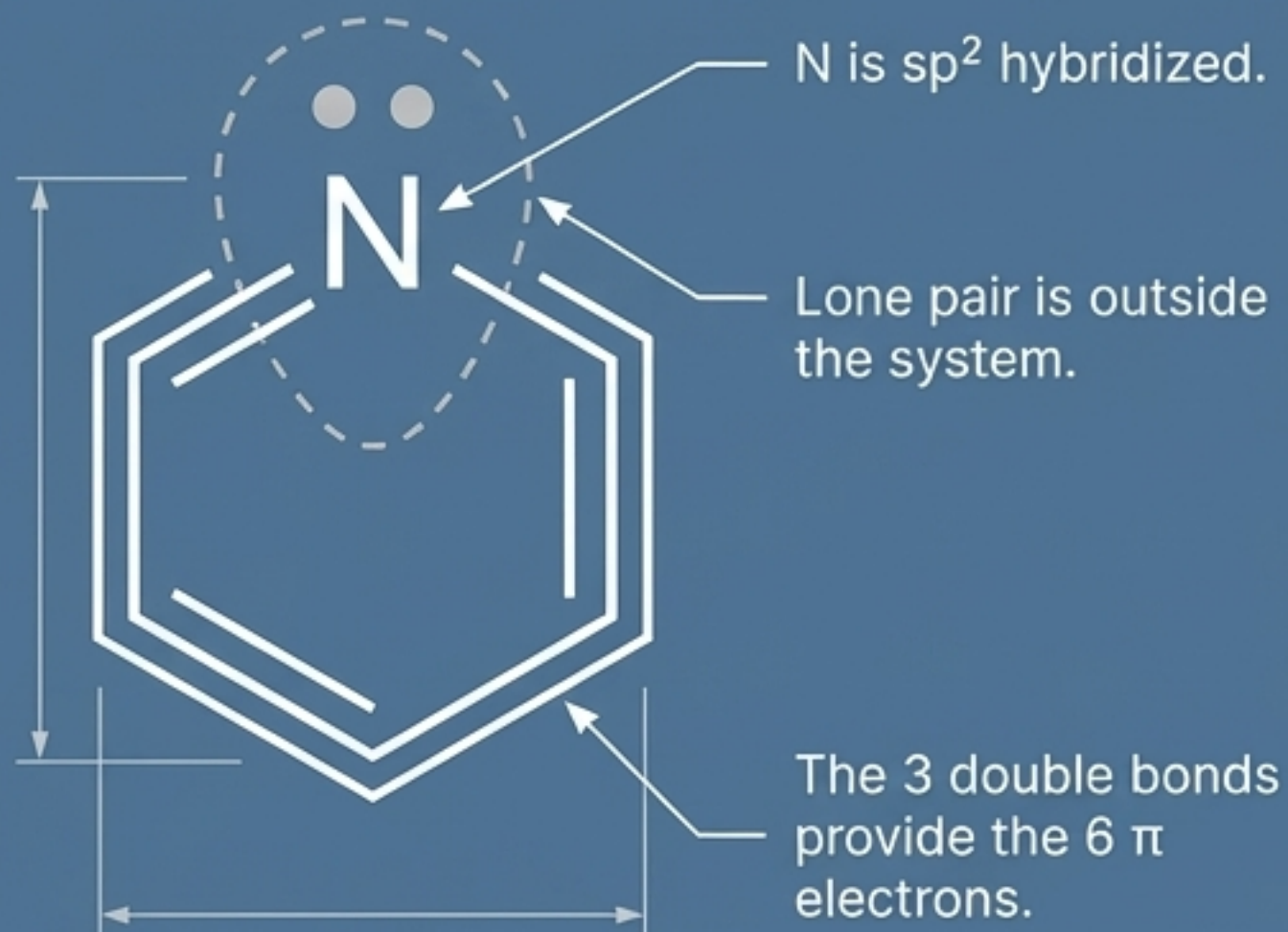
The Trap: **Steric hindrance (hydrogen crowding)** forces the ring to warp out of planar alignment.

Planarity is non-negotiable.
Result: **Non-aromatic.** ✗

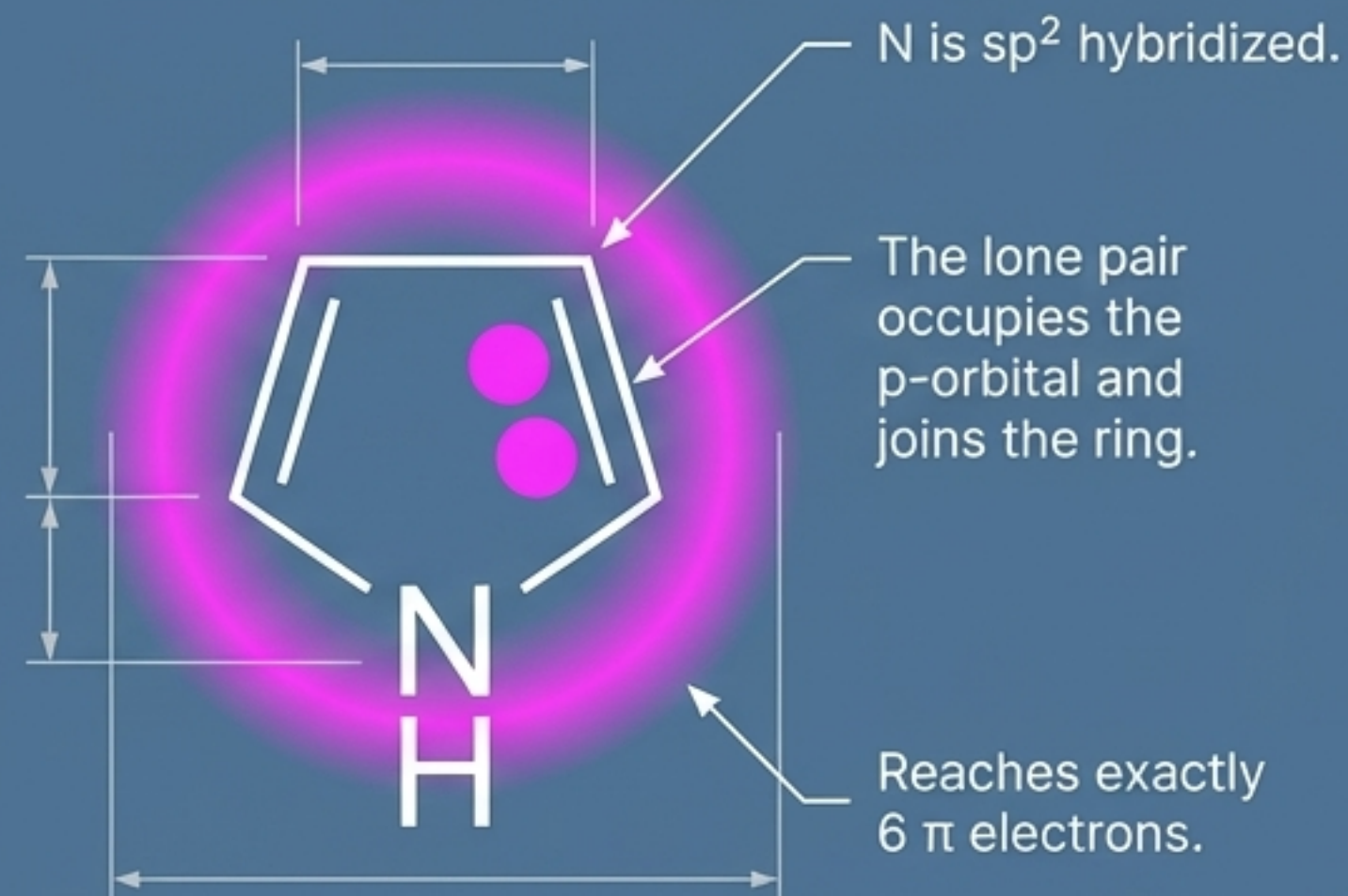
Heterocycles: The Lone Pair Dilemma

Heterocycles contain non-carbon atoms (N, O, S). Do their lone pairs count?

Pyridine

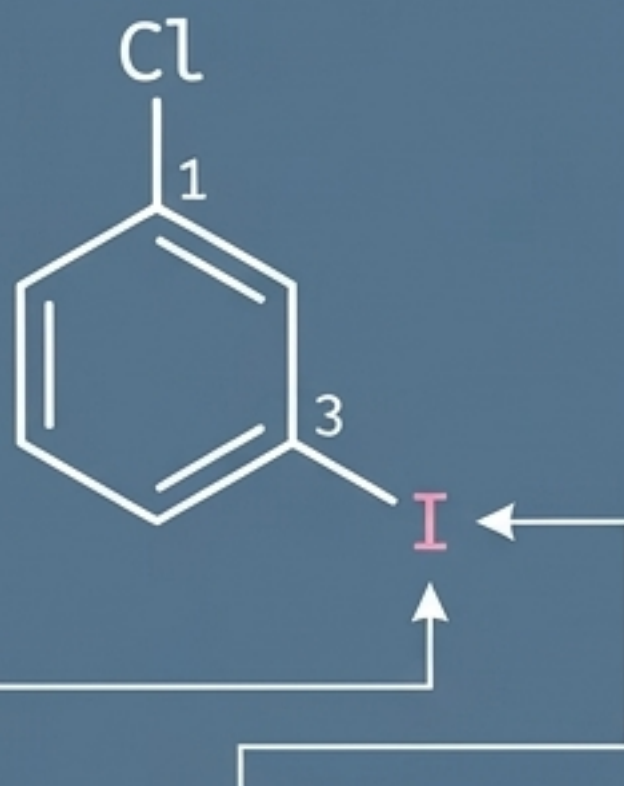


Pyrrole



The Language of Arenes: Basic Nomenclature

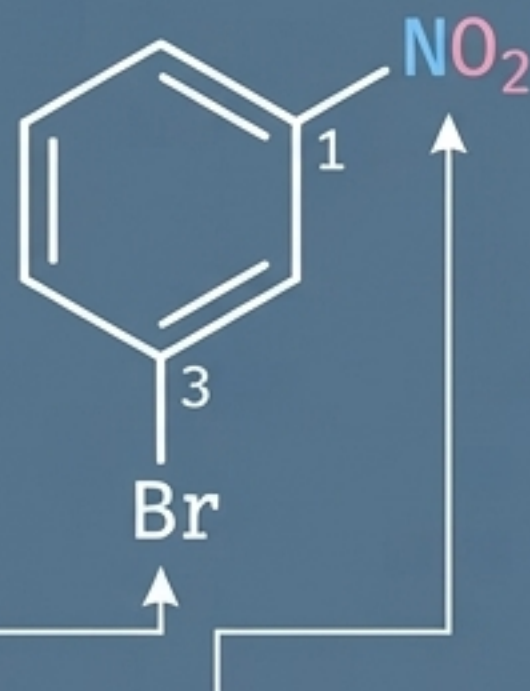
All 6 hydrogens on a bare benzene ring are equivalent.



1-chloro-3-iodobenzene

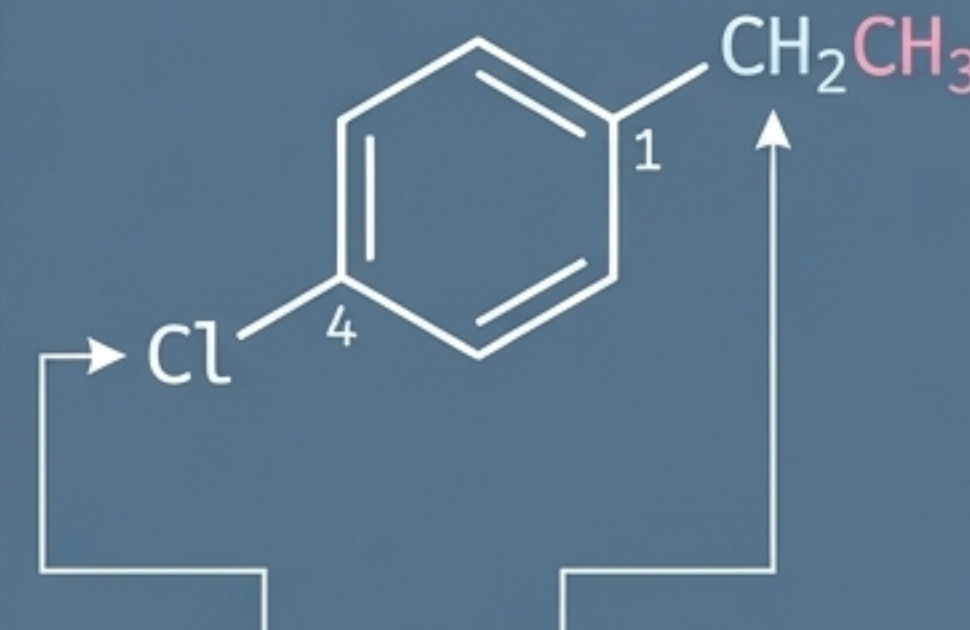
meta-chloriodobenzene

not 1-iodo-3-chlorobenzene or
meta-iodochlorobenzene



1-bromo-3-nitrobenzene

meta-bromonitrobenzene



1-chloro-4-ethylbenzene

para-chloroethylbenzene

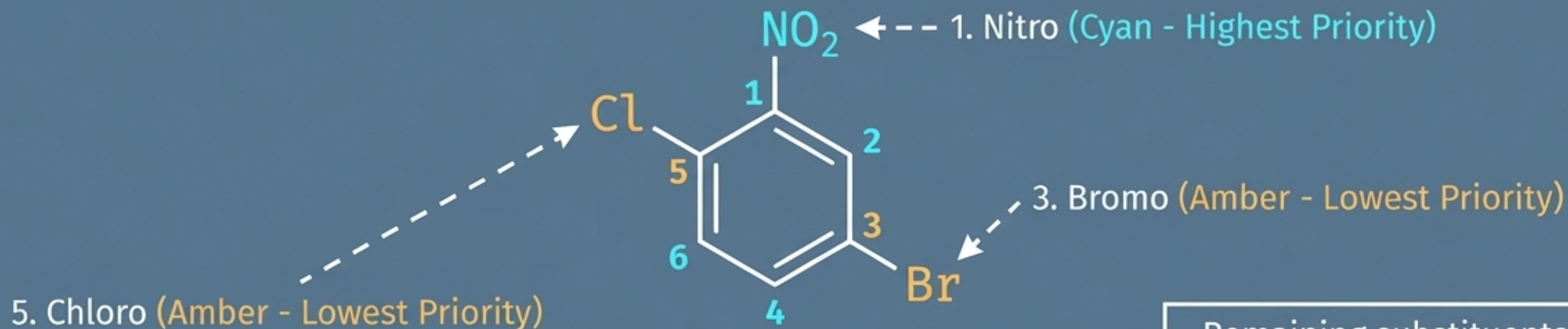
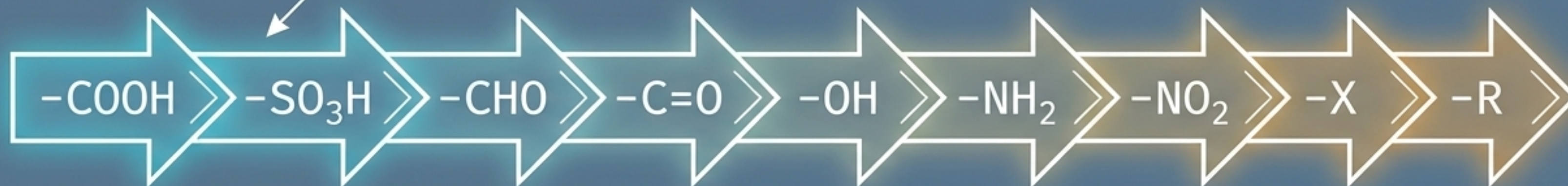
Disubstituted Topography

When two groups are present, exactly three positional isomers exist.



Polysubstituted Priority Hierarchy

The highest priority group dictates position #1.

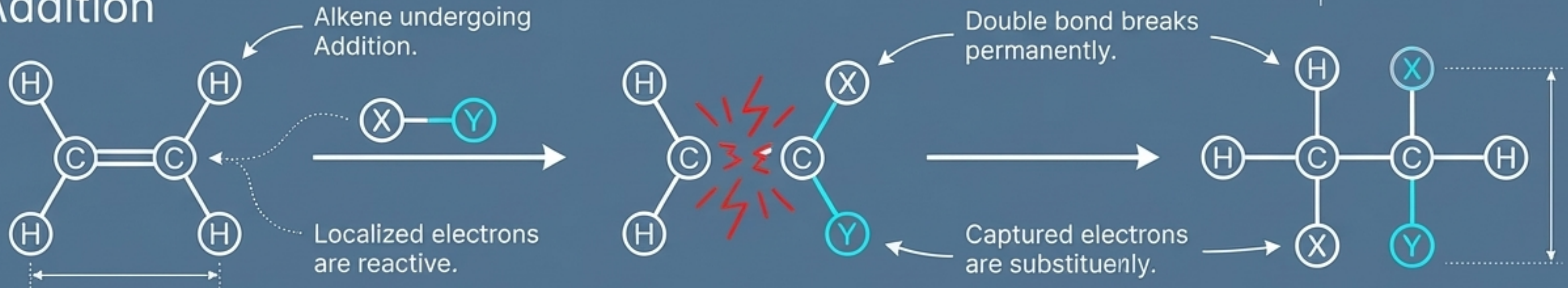


3-Bromo-5-chloronitrobenzene

Remaining substituents are numbered to give the lowest possible positional values and listed alphabetically.

The Consequence of Stability: Substitution over Addition

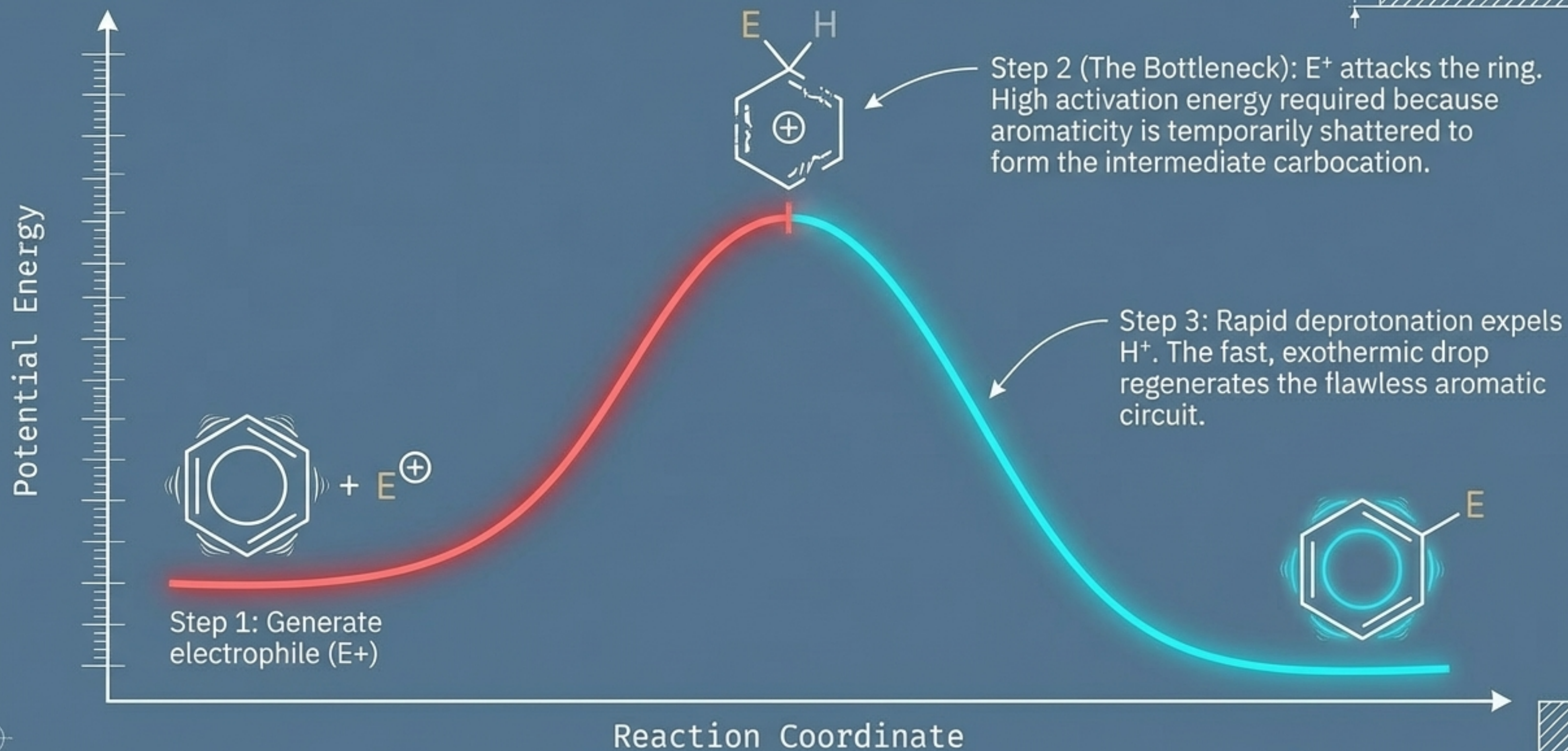
Addition



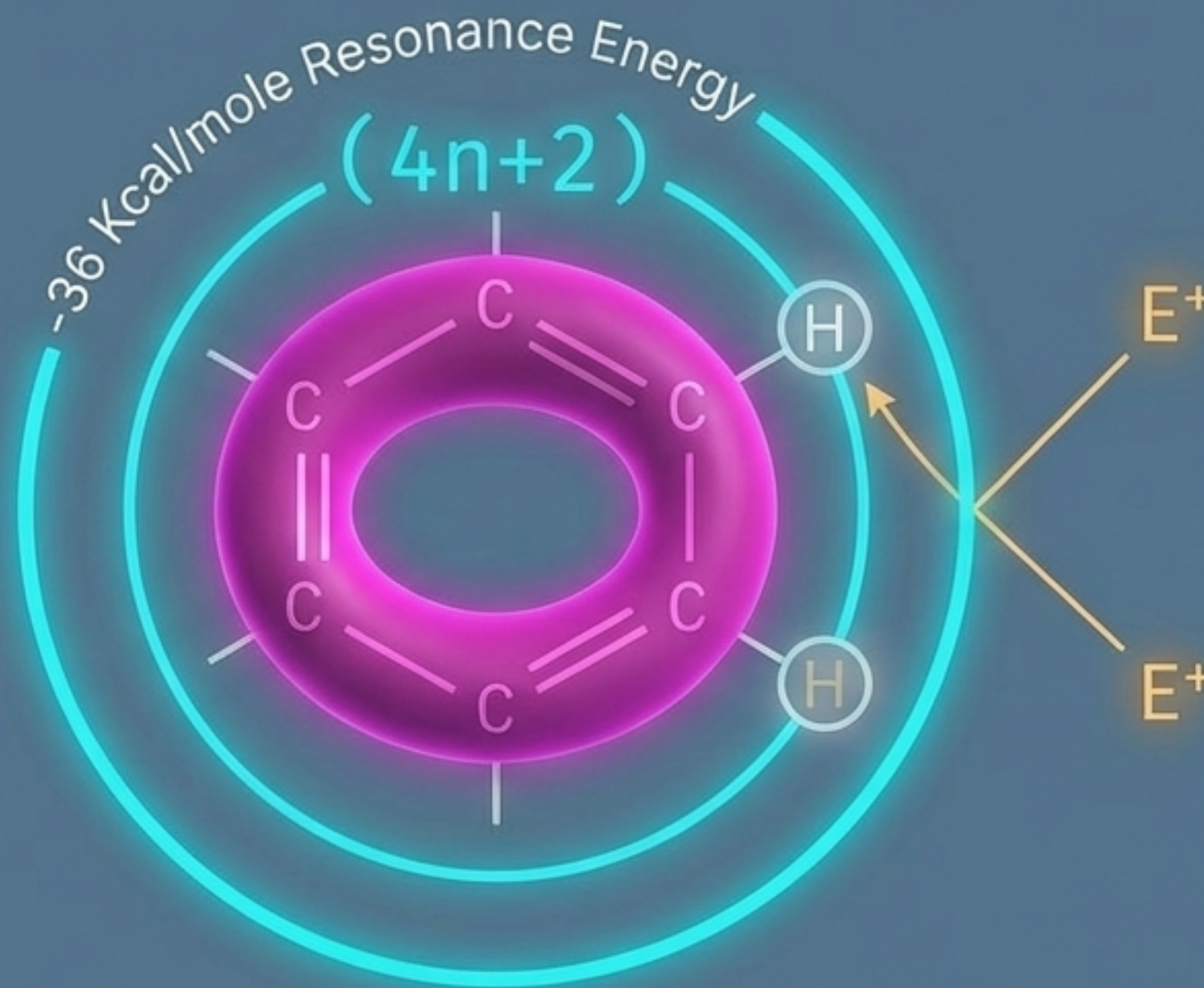
Substitution



The Electrophilic Substitution (SE) Mechanism



Synthesis: The Inviolable Circuit



1. Structure fulfills Math:

Physical delocalization perfectly satisfies the $4n+2$ equation.

2. Math generates Armor:

This specific configuration creates a 36 Kcal/mole thermodynamic shield.

3. Armor dictates Destiny:

The molecule sacrifices outer hydrogens purely to keep its inner pi-circuit unbroken.